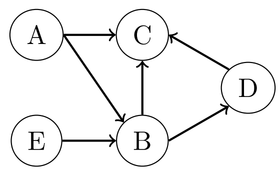
Worksheet 1

Exercise 1: Graph terminology

Have a look at the following causal graphical model.



a) Is the represented graphical model a directed acyclic graph (DAG)? Justify your answer.

It is a DAG, since it is not possible to walk on a circle when following the edges along the direction indicated by the arrows.

b) Give 3 examples of vertices-sequences that correspond to a directed path and undirected paths.

Directed paths: (E,B,D,C), (A,B,C) and (A,B,D)
Undirected paths: (D,B,C,A), (C,B,D) and (E,B,A)
This is not the only possible answer. The graph contains more directed and undirected paths.

c) Determine children, parents and ancestors for the two nodes B and D.

Children:
$Ch(B) = C, D$
$Ch(D) = C$
Parents:
$Pa(B) = A, E$
$Pa(D) = B$
Ancestors:
$An(B) = A, E$
$An(D) = A, E, B$

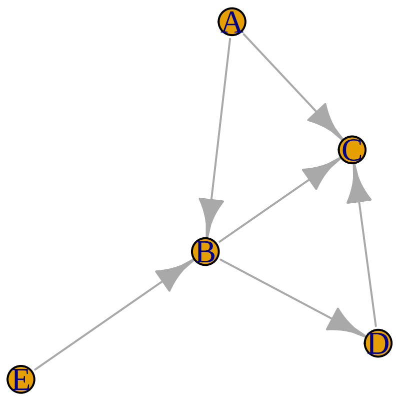
d) Is the total cause of B on C transported by the path B $\rightarrow$ C? Justify your answer.

No, the path B $\rightarrow$ C does only transport the direct cause from B to C. To get the total cause, we need to look at **all** directed paths from B to C, i.e. path B $\rightarrow$ C and path B $\rightarrow$ D $\rightarrow$ C.

e) Draw the graph in R using the function `dag` from the R-Package `gRbase` and repeat c) by using R-Commands.

```
library(gRbase)
# List all nodes with their parents
graphx  <- dag("A",
               c("B","A","E"),
               c("C","A","B","D"),
               c("D","B"),
               c("E"))
```

```r
# Draw graph
plot(graphx)
```




```r
# Children
children("B", graphx)
```

```
[1] "C" "D"
```

```r
children("D", graphx)
```

```
[1] "C"
```

```r
# Parents
parents("B", graphx)
```

```
[1] "A" "E"
```

```r
parents("D", graphx)
```

```
[1] "B"
```

```r
# Ancestors
ancestralSet("B", graphx)
```

```
[1] "A" "B" "E"
```

```r
ancestralSet("D", graphx)
```

```
[1] "A" "B" "E" "D"
```

Exercise 2: Travel time

A new inter-city route A was built between two cities. In order to compare the new route A with the old route B, the travel time in minutes was measured for each route by 250 randomly selected drivers. The data set `reisezeit` in the file `NewRoute.rda` contains the measured times.

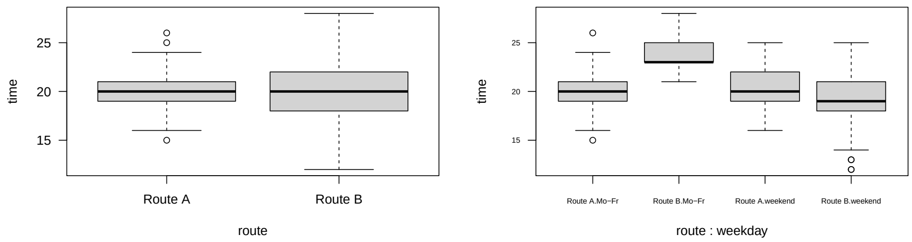
a) Analyze the data and evaluate whether the travel time has been reduced with the new route A or not (time route A $<$ time route B).

```r
load("NewRoute.rda")
```

```r
str(reisezeit)
```

```
'data.frame':	500 obs. of  3 variables:
 $ time   : num  21 17 21 19 19 22 21 12 19 19 ...
 $ route  : chr  "Route A" "Route A" "Route A" "Route A" ...
 $ weekday: Factor w/ 2 levels "Mo-Fr","weekend": 1 1 2 1 2 1 1 2 1 1 ...
```

```r
par(mfrow = c(1,2))
boxplot(time ~ route, data = reisezeit, las = 1)
boxplot(time ~ route + weekday, data = reisezeit, cex.axis = 0.6, las = 1)
```



Without taking into account the day of the week (confounding variable), the mean travel time for route A and route B is practically identical. If you add the day of the week, you can see that route B performs much worse than route A during the week.

```r
aggregate(time ~ route, data = reisezeit, mean)
```

```
    route     time
1 Route A 20.08679
2 Route B 19.84681
```

```r
aggregate(time ~ route + weekday, data = reisezeit, mean)
```

```
    route weekday     time
1 Route A   Mo-Fr 20.02347
2 Route B   Mo-Fr 23.75676
3 Route A weekend 20.34615
4 Route B weekend 19.11616
```

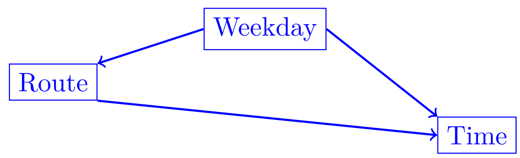
b) Draw a causal diagram visualizing the relation between Time, Route and Weekday.

From a) we know that day of the week is a confounding variable.

```r
table(reisezeit$route, reisezeit$weekday)
```

```
          Mo-Fr weekend
  Route A   213      52
  Route B    37     198
```

Route A is mainly used on weekdays and route B on weekends. The weekday seems to effect the choice of the route. In addition, you can assume that there are more cars on the road on weekdays than on weekends due to work, which potentially leads to longer driving times.



d) What interventions would need to be taken to estimate the causal effect of the route on travel time?

Intervention 1: Everyone must take Route A
Intervention 2: Everyone must take Route B
However, these interventions are hypothetical, since no person can drive on Route A and Route B at the same time or under exactly the same conditions.

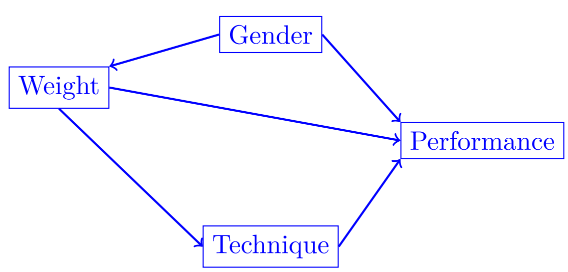
e) How would you design experimental data to measure the effect of the new Route?

To measure the effect of the new Route we need to control for the confounder weekday. The original study needs to be revised by randomly assigning 500 drivers to a route (A or B) and measuring the time on a pre-defined travel day (weekday or weekend).

Exercise 3: Weightlifting

A study aims to analyze the effects of weight and technique on weightlifting performance, which is measured by the number of repetitions. The researchers hypothesize that men tend to select heavier weights due to overconfidence. Additionally, men are generally stronger and therefore able to perform more repetitions. The chosen weight is also believed to influence lifting technique.

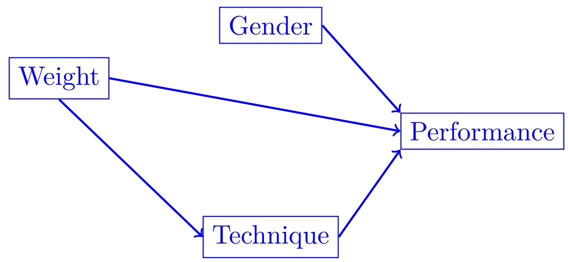
a) Draw a causal diagram of the four variables `technique`, `weight`, `gender` and `performance`.



b) Which intervention would need to be carried out to estimate a causal effect of weight on performance?

Let's choose e.g. 3 specific weight: light, medium, heavy.
3 Interventions: **Everyone** must perform the training with all 3 weights.
The intervention disrupts the confounding factor gender.

c) How does the causal graphical model look like under the intervention b)?



d) Does the causal effect of weight on performance change when all participants receive a specific instruction on the technique?

Yes, the causal effect of weight on performance changes because there is a directed path from weight to performance that includes the "technique" node (indirect effect). The total causal effect of weight on performance is determined by a direct and an indirect effect. The specific instruction changes the indirect effect.

Exercise 4: Oberservational vs. Experimental Data

a) You recruit 300 persons suffering from cold symptoms. You feed all 300 persons all lots of oranges, which are full of vitamin C. Eight days later, 292 out of 300 people are free of cold symptoms. Did the oranges cure people?

Cold symptoms often resolve on their own within a week or two. The recovery observed is most likely due to the natural course of the illness rather than due to the effect of the oranges. Without a control group (persons that suffer from cold symptoms and do not consume oranges) it is impossible to tell whether the oranges had any effect. Moreover, there could be other factors influencing recovery, such as rest, hydration, medications etc. You can conclude that a significant number of people recovered, but we cannot conclude that oranges were the sole cause of the recovery.

b) You recruit 300 persons for a winter race on a test track. You assign randomly 150 persons to cars with winter tires and the other 150 persons to cars with summer tires. There are 30 fewer breakdowns in the group with winter tires. Is driving with winter tires safer?

The effect of summer and winter tires is collected by a randomized controlled experiment. The two groups are comparable since persons are randomly assigned and the test track is the same for everyone. Therefore, we can conclude that driving with winter tires is safer.

c) You are comparing two female teams of floorball players: U21 vs. seniors. There were 8 cases of cruciate ligament injuries in the U21 team last year and only 1 case in the senior team. Can we conclude from this that the senior team is stronger?

It is important to note that injury rates alone do not provide a complete picture of a team's overall strength. Other factors such as training, experience, physical conditioning, and playing style also play a crucial role in determining a team's strength. Moreover, we have two very diverse groups. Therefore, this study could not be used for causal claims.

Exercise 5: Pearl's ladder

Formulate possible research questions on association, intervention and imagination for the following pairs (with reference to Pearl's ladder).

a) Effect of snow and ice on car accidents

Association: What is the expected number of car accidents for a winter with lots of snow?
Intervention: If someone only drives on snow and ice-free days, is he less likely to have a car accident?
Imagination: I didn't have an accident last winter. Was it due to the mild temperatures?

b) Effect of smoking on lung cancer

Association: The percentage of people suffering from lung cancer is higher among smokers.
Intervention: If someone stops smoking today, will his risk of lung cancer be lower?
Imagination: Would my grandmother not have had lung cancer if she would not have smoked?

c) Effect of junk food on obesity

Association: Have people that eat a lot of junk food a higher body weight?
Intervention: If someone stops eating junk food, will he have less body weight?
Imagination: Would my grandfather have weighed less if he hadn't eaten junk food in his life?

d) Effect of wind on speed of sailboat

Association: The speed of a sailboat is higher if there is a lot of wind.
Intervention: If we go sailing on a day without wind, will the speed of the sailboat be very low?
Imagination: Yesterday, we had 1 hour to cross the lake. What would have been the speed of the sailboat if the wind would not have been so strong?